

Part 1

Practice Problems #14 – #16

Column-Row Expansion & Block Matrix | LU Factorization

Permuted LU & Permutation Matrix | Vector Space T/F

Practice Problem #14 — Practice problems #14

Question 1

1. Let

$$\begin{bmatrix} 6 & 4 \\ 2 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 & -1 \\ -1 & -3 & 7 & 1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}, \text{ and } B = \begin{bmatrix} -3 & 7 \\ 1 & 3 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 & -1 \\ -1 & -3 & 7 & 1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

(a) Compute AB using row-column rule.

(b) Compute AB using column-row expansion.

(c) Let A and B be decomposed as follows, where A is of size 2×2 . Compute AB

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

using block matrices.

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 5 \\ -1 & -3 \end{bmatrix}, B = \begin{bmatrix} -2 & 3 \\ 7 & 1 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

Work / Solution

Question 2

2. Suppose that A is an invertible matrix. Find matrices X and Y such that the product

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

below has the form indicated. Also, compute B .

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -2 & 3 \\ -1 & -3 & 7 & 1 \end{bmatrix}$$

$$\begin{bmatrix} I & 0 \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} 0 \\ B_{22} \end{bmatrix}$$

Work / Solution

Question 3

3. Suppose A is invertible. Find X and Y so that

$$\begin{bmatrix} I & 0 \\ A & S \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} 0 \\ B \end{bmatrix}$$

where $S = A - A_{21}A_{11}^{-1}A_{12}$. The matrix S is called the Schur complement of A .

Likewise, if A is invertible, the matrix $A - A_{21}A_{11}^{-1}A_{12}$ is called the Schur complement of A . Such expression occur frequently in the theory of systems engineering,

and elsewhere (don't need to memorize).

Note that this is a factorization of A into block diagonal and block triangular matrices.

Work / Solution

Practice Problem #15 — Practice problems #15

Question 1

$$\begin{bmatrix} 1 & 2 & 4 & 0 \\ 2 & -2 & 4 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Work / Solution

Question 2

1. Let $A = \begin{bmatrix} 1 & -3 & 1 \\ 3 & 7 & 5 \\ 7 & 5 & 7 \end{bmatrix}$ and $b = \begin{bmatrix} -5 \\ 3 \\ 7 \end{bmatrix}$.

$$\begin{bmatrix} 1 & -3 & 1 \\ 3 & 7 & 5 \\ 7 & 5 & 7 \end{bmatrix}$$

$$\begin{bmatrix} -5 \\ 3 \\ 7 \end{bmatrix}$$

(a) Find an LU factorization of A.

(b) Use (a) to solve the equation $Ax = b$.

Work / Solution

Question 3

2. True or False. Justify your answer:

Suppose a matrix A is row reduced to an echelon form U. If $A = LU$ is an LU factorization, then A can be row reduced to U using only replacement operations.

(This is asking the converse of what we did in class. In class, we showed that if A is row reduced to U using only replacement operations, then $A = LU$.)

Answers or Hints on page 2.

Practice Problem #16 — Practice problems #16

Question 1

1. Suppose $PA = LU$ where P is a permutation matrix. Explain the steps to solve the equation $Ax = b$.

$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 6 \\ 1 & 1 & 0 \end{bmatrix}$

$\begin{bmatrix} 0 & 0 & 6 \\ 1 & 1 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 6 \end{bmatrix}$

Work / Solution

Question 2

2. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 2 & 1 \end{bmatrix}$.

$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 2 & 1 \end{bmatrix}$

$\begin{bmatrix} 0 & 4 & 5 \\ 1 & 2 & 1 \end{bmatrix}$

(a) Give a permutation matrix P so that PA is upper triangular.

(b) Give two permutation matrices P and Q so that PAQ is lower triangular.

$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}$

Work / Solution

Question 3

3. Find a 3×3 permutation matrix P such that $P^3 = I$.

Work / Solution

Question 4

4. True or False. Justify.

(a) Suppose V is a set of all 2×2 real matrices with nonnegative entries. With the sum and scalar multiplications defined as usual (that is, elementwise, as we know of), V is a vector space.

(b) Suppose V is a set of 2×2 symmetric matrices. With the sum and scalar multiplications defined as usual (that is, elementwise, as we know of), V is a vector space.

Answers are on the back.

Work / Solution

Answer Key / Hints — Part 1

■ Practice Problem #14

#1: Do them! You should get same answer for all.

#2: $X = -A A^{-1}$, $Y = -A A^{-1}$, $B = A -A A^{-1}A$

21 11 31 11 22 22 21 11 12

#3: $X = A A^{-1}$, $Y = A -1A$

21 11 11 12

■ Practice Problem #15

$\begin{bmatrix} 1 & 0 & 0 & 2 & -2 & 4 & 0 & -5 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

1 0 0 2 -2 4 0 -5

$\begin{bmatrix} 1 & 0 & 0 & 2 & -2 & 4 & 0 & -5 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

#1: $L = \begin{bmatrix} 1 & 1/2 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$, $U = \begin{bmatrix} 0 & -2 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $y = \begin{bmatrix} -5 \\ 0 \\ 0 \end{bmatrix}$, $x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 1/2 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

3/2 -5 1 0 0 -6 -18 3

#2: True. (Since L means a unit lower triangular matrix in the LU factorization, L can be reduced to I using only which row operation? So, $E \cdots E L = I$ where E are what kind of $k \ 1 \ i$

elementary matrices? Then, $A = LU$ becomes $E \cdots E A = E \cdots E LU$.)

$k \ 1 \ k \ 1$

■ Practice Problem #16

#1: Premultiply P to the equation to get $P Ax = P b$ (that is, permute entries of b) which then becomes $L U x = b'$, ($b' = P b$). Solve this equation as usual LU .

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

0 1 0

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

#2(a): $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

1 0 0

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

1 0 0 0 0 1

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

#2(b): $P \ 1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, $P \ 2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

0 1 0 1 0 0

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

0 1 0

#3: $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ or P^T .

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

1 0 0

#4(a): False. Hint: closed under scalar multiplication? counter-example? Note that to be a vector space, the scalar multiplication thing should be true for all scalars.

#4(b): True. Sum of two symmetric matrices symmetric? How about scalar multiple? Is there zero vector? (Is zero matrix symmetric?) Is additive inverse of a symmetric matrix symmetric? Must show all (10) axioms are satisfied.

Part 2

Practice Problems #17

Subspace (Vector Space Axioms & Proofs)

Practice Problem #17 — Practice Problems #17

Question 1

1. True or False. Justify your answer.

Let $V = \mathbb{R}^2$, which we showed is a vector space with the usual operations

$$u+v = (u_1+v_1, u_2+v_2), \quad cu = (cu_1, cu_2).$$

$$H = \{(x, y) : x \leq y\}$$

Then,

$$H = \{(x, y) : x \leq y\},$$

is a subspace of V .

Work / Solution

Question 2

2. Show $c0 = 0$ for every scalar c following the proof below. Justify each step.

$$\text{Proof: } c0 = c(0 + 0) = + \quad .$$

Then,

$$0 = -c0 + c0 = (-c0 + c0) + = 0 + c0 = \quad .$$

Work / Solution

Question 3

3. For $u \in V$, where V is a vector space, suppose

$$\alpha u = 0$$

for some nonzero scalar α . Show that $u = 0$. Justify each and every step in your proof.

Answer Key / Hints — Part 2

■ Practice Problem #17

#1: False. Why?

#2: Mimic what we did in class.

#3: Note that $u = 1u$ (which axiom?). Use the result from #2.

Part 3

Practice Problems #18 – #26

Determinants | Determinant Properties | Cramer's Rule & Adjugate
Area, Volume & Linear Transformations | Col A & Ker T | Bases
Coordinate Mapping | Dimension | Change of Basis

Practice Problem #18 — Practice problems #18

Question 1

1. We know that $\mathbb{R}^{2 \times 2}$ is a vector space with usual element-wise addition and scalar multiplication.

$$\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$$

(a) Show that the set H of all matrices of the form is a subspace of $\mathbb{R}^{2 \times 2}$.

$$\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$$

(b) Give a spanning set for H .

Work / Solution

Question 2

2. Compute the determinant of the matrix A using

(a) cofactor expansion

(b) short-cut formula given in class.

$$\begin{bmatrix} 1 & -2 & 7 \\ -4 & 8 & 5 \end{bmatrix}$$

$A = \begin{bmatrix} 1 & -2 & 7 \\ -4 & 8 & 5 \end{bmatrix}$.

$$\begin{bmatrix} 1 & -2 & 7 \\ -4 & 8 & 5 \end{bmatrix}$$

Work / Solution

Question 3

3. Compute the determinant.

$$\begin{bmatrix} 1 & 0 & 0 & 3 \\ 1 & 0 & 0 & 3 \end{bmatrix}$$

| 2 7 0 6 |

A = | |

| |

| 0 6 3 0 |

[]

7 3 1 -5

Answers or Hints on page 2.

Work / Solution

Practice Problem #19 — Practice problems #19

Question 1

1. Find the determinant by row reduction to echelon form:

[]

1 3 -1 0 -2

| |

| 0 2 -4 -1 -6 |

| |

| |

A = | -2 -6 2 3 9 |

| |

| |

| 3 7 -3 8 -7 |

[]

3 5 5 2 7

Work / Solution

Question 2

2. Combine the methods of row reduction and cofactor expansion to compute the determinant:

[]

-3 -2 1 -4

| |

| 1 3 0 -3 |

A = | |

| |

| -3 4 -2 8 |

[]

3 -4 0 4

Work / Solution

Question 3

3. Let A and B be 4×4 matrices, with $\det A = -1$ and $\det B = 2$. Compute:

- (a) $\det AB$
- (b) $\det (B^5)$
- (c) $\det 2A$

Work / Solution

Question 4

4. True or False. Justify your answer.

- (a) Let A and B be square matrices, with B invertible. Then, $\det(BAB^{-1}) = \det A$.
- (b) Suppose that A is a square matrix such that $\det(A^4) = 0$. Then, columns of A are linearly independent.
- (c) Let U be a square matrix such that $UU^T = I$. Then $\det U = \pm 1$.

Answers or Hints on page 2.

Work / Solution

Practice Problem #20 — Practice problems #20

Question 1

1. Determine the values of the parameter s for which the given system has a unique solution, and then describe the solution using Cramer's Rule.

$$\begin{cases} s x_1 + 2s x_2 = -1 \\ 3x_1 + 6s x_2 = 4 \end{cases}$$

Work / Solution

Question 2

2. Compute the adjugate of the given matrix, and the inverse of the matrix in terms of adjugate.

$$\begin{bmatrix} 3 & 6 & 7 \\ 0 & 2 & 1 \\ 2 & 3 & 4 \end{bmatrix}$$

Work / Solution

Question 3

3. True or False. Justify your answer.

Suppose that all the entries in A are integers and $\det A = 1$. Then, all the entries in A^{-1} are integers.

Answers or Hints on page 2.

Practice Problem #21 — Practice problems #21

Question 1

1. Find the volume of the parallelepiped with one vertex at the origin and adjacent vertices at $(1, 4, 0)$, $(-2, -5, 2)$, $(-1, 2, -1)$.

Work / Solution

Question 2

2. Let R be the triangle with vertices at (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) . Show that area of R is
- $$\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$
- the triangle R equals $\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$.

Work / Solution

Question 3

3. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation that maps e_1, e_2, e_3 to $ae_1 + be_2 + ce_3$, $e_1 + e_2 + e_3$, $e_1 + e_2$ respectively, where a, b and c are some positive real numbers. Let S be the unit ball in $\mathbb{R}^3$ bounded by the unit sphere $x^2 + y^2 + z^2 = 1$. Use the fact that the volume of the unit ball is $\frac{4\pi}{3}$ to find the volume of $T(S)$.

Practice Problem #22 — Practice problems #22

Question 1

1. Find A such that the given set is Col A:

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} \right\}$$

$$b - c$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$|$$

$$| \quad 2b + c + d$$

$$|$$

$$|$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$| \quad | : b, c, d \text{ real } .$$

$$| \quad |$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$|$$

$$5c -$$

$$d$$

$$4d \quad |$$

$$|$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

Work / Solution

Question 2

2. Let $T : V \rightarrow W$ be a linear transformation from a vector space V into a vector space W . Prove that $\text{Range } T$ is a subspace of W .

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$p(0)$$

Work / Solution

Question 3

3. Define a linear transformation $T : P \rightarrow R^2$ by $T(p) =$. For instance, if

2 L J

 $p(1)$

[]

3

$p(t) = 3+5t+7t^2$, then $T(p) =$.

LJ

15

(a) Show that T is a linear transformation.

(b) Find a polynomial $p \in P$ that spans the Kernel of T , i.e., find a spanning set for

2

 $\text{Ker } T.$

(c) Describe the range of T .

Answers are on page 2.

Practice Problem #23 — Practice problems #4

Question 1

1. True or False. Justify your answer:

(a) The set $B = \{1+t^2, t+t^2, 1+2t+t^2\}$ in P is linearly independent.

2

{ }

1 2 5 11 3 0 2 5 11 3

■ ■ ■ ■

82 2 4 5 15 2 3 2 2 4 5 15 2 39

(b) The set $B = \{>>>><\}$

6

6

1 2

■

0 4 5 7

7

,

6

6

1 2

■

0 4 5 7

7

$>>>>=$ in R_4 ■ 5 is lin-

6 7 6 7

6 3 6 5 19 2 7 6 3 6 5 19 2 7

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4nt.

■ ■ 7

5

6

4

■ ■ 7

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Work / Solution

Question 2

2. Find a basis for the set of vectors (x, y, z) in $\mathbb{R}^3$ such that $x+2y+z=0$.

Work / Solution

Question 3

3. Find bases for $\text{Nul } A$ and $\text{Col } A$, where A is row equivalent to B :

$\begin{bmatrix} 1 & 2 & 5 & 11 & 3 & 1 & 2 & 0 & 4 & 5 \end{bmatrix}$

$\begin{bmatrix} \blacksquare & \blacksquare \end{bmatrix}$

$\begin{bmatrix} 2 & 2 & 4 & 5 & 15 & 2 & 3 & 2 & 0 & 0 & 5 & 7 & 8 & 3 \end{bmatrix}$

$A = \begin{bmatrix} \blacksquare \end{bmatrix}$, $B = \begin{bmatrix} \blacksquare \end{bmatrix}$.

$\begin{bmatrix} 6 & 1 & 2 & 0 & 4 & 5 & 7 & 6 & 0 & 0 & 0 & 0 & 9 & 7 \end{bmatrix}$

$\begin{bmatrix} 6 & 7 & 6 & \blacksquare & 7 \end{bmatrix}$

$\begin{bmatrix} 6 & 7 & 6 & 7 \end{bmatrix}$

$\begin{bmatrix} 6 & 3 & 6 & 5 & 19 & 2 & 7 & 6 & 0 & 0 & 0 & 0 & 0 & 7 \end{bmatrix}$

$\begin{bmatrix} 6 & \blacksquare & \blacksquare & 7 & 6 & 7 \end{bmatrix}$

$\begin{bmatrix} 4 & 5 & 4 & 5 \end{bmatrix}$

$\begin{bmatrix} 4 & 1 & 7 \end{bmatrix}$

Work / Solution

Question 4

4. Let $v_1 = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix}$, $v_2 = \begin{pmatrix} 2 \\ 9 \\ 3 \end{pmatrix}$, $v_3 = \begin{pmatrix} 2 \\ 11 \\ 3 \end{pmatrix}$, and $H = \text{Span } v_1, v_2, v_3$. It can be
 $\blacksquare \{ \}$
 $\begin{pmatrix} 6 & 7 & 7 & 6 & 2 & 7 & 6 & 6 & 7 \end{pmatrix}$
 $\begin{pmatrix} 6 & 7 & 6 & \blacksquare & 7 & 6 & 7 \end{pmatrix}$
 verified that $4v_1 + 5v_2 + 4v_3 = 50$. Use this information to find a basis for H .
 $\begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$
 $\blacksquare$

Work / Solution

Question 5

5. True or False. Justify:

(a) Suppose $v_1, \dots, v_n$ are vectors in $\mathbb{R}^4$ such that $\mathbb{R}^4 = \text{Span } v_1, \dots, v_n$. Then

$\begin{pmatrix} 1 & 4 & 1 & 4 \end{pmatrix}$

$\dots \{ \dots \}$

$v_1, \dots, v_n$ is a basis for $\mathbb{R}^4$.

$\begin{pmatrix} 1 & 4 \end{pmatrix}$

$\{ \dots \}$

(b) Let $B = v_1, \dots, v_n$ be a linearly independent set in $\mathbb{R}^n$. Then B is a basis for

$\begin{pmatrix} 1 & n \end{pmatrix}$

$\{ \dots \}$

$\mathbb{R}^n$.

(c) Every vector space has a basis.

Answers are on the back.

Work / Solution

Practice Problem #24 — Practice problems #24

Question 1

1. Let V be a vector space with a basis $B = \{b_1, b_2, \dots, b_n\}$. Show that the coordinate mapping $T : V \rightarrow \mathbb{R}^n$ defined by $T(x) = [x]_B$ satisfies
- $$T(cx) = cT(x)$$
- for all $x \in V$ and for all scalars c .

Work / Solution

Question 2

2. Find the vector x determined by the given coordinate vector $[x]_B$ and the given basis $B = \{b_1, b_2, b_3\}$.
- $$[x]_B = \begin{bmatrix} 1 \\ 5 \\ 4 \\ 3 \end{bmatrix}, \quad b_1 = \begin{bmatrix} -4 \\ 2 \\ -7 \\ 0 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 2 \\ 2 \\ -7 \\ 0 \end{bmatrix}, \quad b_3 = \begin{bmatrix} -7 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Work / Solution

Question 3

3. Prove that the set $B = \{1+t^2, t+t^2, 1+2t+t^2\}$ is a basis for P_2 .

Practice Problem #25 — Practice Problems #25

Question 1

1. In the following, V is a nonzero finite-dimensional vector space, and the vectors listed belong to V .

True or False. Justify:

(a) If there exists a set $\{v_1, \dots, v_p\}$ that spans V , then $\dim V \leq p$.

1 p

(b) If there exists a linearly independent set $\{v_1, \dots, v_p\}$ in V , then $\dim V \geq p$.

1 p

(c) If $\dim V = p$, then there exists a spanning set of $p+1$ vectors in V .

(d) If there exists a linearly dependent set $\{v_1, \dots, v_p\}$ in V , then $\dim V \leq p$.

1 p

Work / Solution

Question 2

2. Prove that the first four Hermite polynomials $1, 2t, -2+4t^2$, and $-12t+8t^3$ are linearly independent in P using Coordinate mapping idea. Then, explain why the set

3

$S = \{1, 2t, -2+4t^2, -12t+8t^3\}$ must be a basis of P .

3

Work / Solution

Practice Problem #26 — Practice problems #26

Question 1

1. In P_2 , find the change-of-coordinates matrix from the basis

B

$B = \{1-2t+t^2, 3-5t+4t^2, 2t+3t^2\}$ to the standard basis $S = \{1, t, t^2\}$.

If $p(t) = -1+2t$, what is the S -coordinate vector for $p(t)$?

Then find the B -coordinate vector for $p(t)$ using the change-of-coordinates matrix.

Work / Solution

Question 2

2. In P_1 , the following B and C are two bases. Find the change-of-coordinates matrix

from B to C ,

and the change-of-coordinates matrix from C to B .

$B = \{7+5t, -3-t\}$, $C = \{1-5t, -2+2t\}$

Hints or Answers are on page 2.

Work / Solution

Answer Key / Hints — Part 3

■ Practice Problem #18

#1(a): Check 4 items as we did in class.

#1(b): $H = \text{span}$ of what vectors? (Here, vectors are matrices.) One possible answer is

$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

#2: 0 for both

#3: -546

■ Practice Problem #19

#1: 24

#2: 0

#3: (a) -2, (b) 32, (c) -16

#4: (a) T, (b) F (c) T. Justify!

■ Practice Problem #20

#1: $s = 0, 1$; $x = -7$, $x = 4s+3$.

$\begin{bmatrix} 1 & 3(s-1) & 2 & 6s(s-1) \end{bmatrix}$

$\begin{bmatrix} 1 & 3 & 2 & 6 \end{bmatrix}$

$\begin{bmatrix} 5 & -3 & -8 & 5 & -3 & -8 \end{bmatrix}$

#2: $\text{adj}A = \begin{bmatrix} 1 & 2 & -2 & -3 \end{bmatrix}$, $A^{-1} = (-1) \begin{bmatrix} 1 & 2 & -2 & -3 \end{bmatrix}$.

$\begin{bmatrix} 1 & 2 & -2 & -3 \end{bmatrix}$

$\begin{bmatrix} -4 & 3 & 6 & -4 & 3 & 6 \end{bmatrix}$

#3: True. (Hint: True for 2×2 case? How about in..... proof?)

■ Practice Problem #21

#1: 15

$\begin{bmatrix} 1 & 1 \end{bmatrix}$

$(\text{cid:12}) \begin{bmatrix} 1 & 1 \end{bmatrix} (\text{cid:12}) x y 1$

$(\text{cid:12}) x -x x -x (\text{cid:12}) 1 1$

#2: The area of R is $\frac{1}{2} (\text{cid:12}) (\text{cid:12}) (\text{cid:12}) \det \begin{bmatrix} y & 1 & -y & 3 & y & 2 & -y & 3 \end{bmatrix} (\text{cid:12}) (\text{cid:12})$

(cid:12) , and the matrix $\begin{bmatrix} 1 & 3 & 2 & 3 \end{bmatrix}$ is row

$(\text{cid:12}) 1 3 2 3 (\text{cid:12})$

$x y 1$

$3 3$

$\begin{bmatrix} 1 & 1 \end{bmatrix}$

$x -x y -y 0$

$1 3 1 3$

$\begin{bmatrix} 1 & 1 \end{bmatrix}$

equivalent to the matrix $\begin{bmatrix} x & 2 & -x & 3 & y & 2 & -y & 3 & 0 \end{bmatrix}$.

$\begin{bmatrix} 1 & 1 \end{bmatrix}$

$x y 1$

$3 3$

#3: $abc4\pi/3$

■ Practice Problem #22

$\begin{bmatrix} 1 & 1 \end{bmatrix}$

$1 -1 0$

```

| |
| 2 1 1 |
#1: | |
| |
| 0 5 -4 |
| |
0 0 1

```

#2: Note that $\text{Range } T = \{b \in W : T(x) = b \text{ for some } x \in V\}$.

$\text{Range } T \subseteq W$ is clear.

The zero vector of W is in range of T because $T(0) = 0$ (since T is linear).

If b, b are in $\text{Range } T$ (which means there are x and x in V such that $T(x) = b$ and

```
1 2 1 2 1 1
```

$T(x) = b$), then $b + b \in \text{Range } T$ because $T(x) = b + b$ has a solution (what's the

```
2 2 1 2 1 2
```

solution?)

Show similarly that if $b \in \text{Range } T$ and c is any scalar, then $T(x) = cb$ has a solution (what's the solution?) so that $cb \in \text{Range } T$.

#3(a): Let p, q be polynomials in P , then

```
[ ] [ ] [ ] [ ] [ ] [ ]
```

$(p+q)(0) \quad p(0)+q(0) \quad p(0) \quad q(0)$

$T(p+q) = = = + = T(p)+T(q)$.

```
[ ] [ ] [ ] [ ] [ ] [ ]
```

$(p+q)(1) \quad p(1)+q(1) \quad p(1) \quad q(1)$

Similarly, show that $T(cp) = cT(p)$.

#3(b): $\text{Ker } T = \{p \in P : p(0) = 0 \text{ and } p(1) = 0\} = \{\alpha t(t-1) : \alpha \in R\}$, i.e.,

$\text{Ker } T = \text{Span } \{t(t-1)\}$. So, the spanning set for $\text{Ker } T$ is $\{t(t-1)\}$.

```
[ ]
```

α

#3(c): Given any vector $\in R^2$, we can form a polynomial p so that $p(0) = \alpha$ and

```
[ ]
```

β

$p(1) = \beta$. For instance, $p(t) = (\beta - \alpha)t + \alpha$ is such polynomial.

Hence $\text{Range } T = R^2$.

■ Practice Problem #23

Answers:

#1(a): True. Consider the equation $c(1+t^2) + c(t+t^2) + c(1+2t+t^2) = 0$.

```
1 2 3
```

#1(b): True. Justify!

#2: This problem is essentially asking to find a basis for $\text{Nul } A$ for an appropriate matrix

```
2 1
```

```
■ ■
```

A. One possible answer: $\begin{bmatrix} 8 & 2 & 1 & 3 \\ 2 & 0 & 3 & 9 \end{bmatrix}$

```
>><6
```

```
0 7 6 1 7
```

```
>>=
```

```
6 7 6 7
```

```
>>: 4 2 5 4 4 5>>;
```

```
■ ■
```

```
8 2 1 3 2 0 3 9
```

#3: Basis for $\text{Nul } A$:

```
>>>>>>>><
```

```
, 6
```



```

7/5
>>>>>>=
6 7 6 7
>>>>>>: 6 6
0 7 7
6 6
0 7 7
>>>>>>;
■ ■
82 2 3 2 5 3 2 2 39
Basis for Col A: >>>><
1 7
,
■
0 7
,
5 7
>>>>=
6 7 6 7 6 7
6 3 7 6 5 7 6 2 7
#4:
{
v
,v
}
or
{
v
>>>>:1 6
4,v
}
5or
{
v
■
,v
}
a
4re
■
all
5p
>>>>;
ossible answers.
#5(a): True. Let  $A = [v \ v]$ . Then,  $A$  is a  $4 \times 4$  matrix. If  $R^4 = \text{Span } v, \dots, v$ , what
1 4 1 4
... ■ { ... }
can you say about the columns of  $A$ ?
#5(b): True. Note that the matrix  $A = [v \ v]$  is  $n \times n$ .
1 n
... ■
#5(c): False. Give an example of a vector space with no basis.

```

■ Practice Problem #24

#1: Mimic what we did in class.

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

-1

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

#2: $\begin{bmatrix} 1 & -5 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

#3: Show two things!

■ Practice Problem #25

#1(a) T, (b) T, (c) T (justify them!), (d) F (create a counterexample)

#2: Mimic what we did in class. What is $\dim P$? How many vectors in S ?

■ Practice Problem #26

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

1 3 0

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

#1: $M_{S \leftarrow B} = \begin{bmatrix} [b_1]_S & [b_2]_S & [b_3]_S \end{bmatrix} = \begin{bmatrix} 1 & -2 & -5 & 2 \end{bmatrix}$. What is $[p(t)]_S$?

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

1 4 3

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$[p(t)]_B = \begin{bmatrix} 1 & -2 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

#2: Using isomorphism idea from P onto R^2 , that is, by using coordinate mapping relative to the standard basis $S = \{1, t\}$, it would be better to write the bases B and C as set of vectors in R^2 . Then, you can apply the algorithm that we did in class to find the change of coordinate matrix M .

$C \leftarrow B$

Ans:

$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

-3 1 -2 1

$M =$, and $M =$

$C \leftarrow B \begin{bmatrix} 1 \\ 1 \end{bmatrix} B \leftarrow C \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

-5 2 -5 3

Note that #1 is easier with definitions, #2 is easier with algorithms.

Part 4

Practice Problems #27

Row Space

Practice Problem #27 — Practice problems #27

Question 1

1. Find bases for RowA and RowB, where A is row equivalent to B:

$\begin{bmatrix} 1 & 2 & -5 & 11 & -3 & 1 & 2 & 0 & 4 & 5 \\ 1 & 2 & 4 & -5 & 15 & 2 & 0 & 0 & 5 & -7 & 8 \end{bmatrix}$

$A = \begin{bmatrix} 1 & 2 & -5 & 11 & -3 & 1 & 2 & 0 & 4 & 5 \\ 1 & 2 & 4 & -5 & 15 & 2 & 0 & 0 & 5 & -7 & 8 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 0 & 4 & 5 & 0 & 0 & 0 & 0 & -9 \\ 3 & 6 & -5 & 19 & -2 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Work / Solution

Question 2

2. Suppose that a nonhomogeneous system of six linear equations in eight unknowns has a solution, with two free variables. Is it possible to change some constants on the equations' right-hand-sides to make the new system inconsistent? Explain.

Work / Solution

Question 3

3. Let A be an $m \times n$ matrix. Which of the subspaces

Row A, Col A, Nul A, Row A^T , Col A^T , and Nul A^T are in $\mathbb{R}^m$ and which are in $\mathbb{R}^n$?

How many distinct subspaces are in this list?

Work / Solution

Question 4

4. Let A be an $m \times n$ matrix. Prove that the equation $Ax = b$ has a solution for all $b \in \mathbb{R}^m$ if and only if the equation $ATx = 0$ has only the trivial solution.

Work / Solution

Answer Key / Hints — Part 4

■ Practice Problem #27

#1: Easy. One answer works for both RowA and RowB.

#2: No. (nullity = 2, so rank = 6, hence??)

#3: Subspaces of R_n are Row A and Nul A; Subspaces of R_m are Col A and Nul A^T ; four distinct subspaces.

#4: Hint: rank A^T = dim Col A^T (by definition) = dim RowA (showed in class)